

Exact Organization of Density Matrices and Entanglement Structure in the Kitaev Spin Liquid

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- Entanglement analysis: Could we construct a reduced density matrix without relying on the fermionic representation?
 - Fermionic representation needs Gutzwiller projection.
 - Use physical **spin** operators to describe a spin system.
- Would spin representation be a better tool to understand the entanglement?
- A recent work on the entanglement of $\mathbb{Z}_2$ gauge theory with 1-form symmetries W.-T. Xu *et al.* PRX **15**, 011001 (2025): Could we apply this theory to make the entanglement structure of the Kitaev model clearer?
 - Directly solve the entanglement spectrum
 - Do not use the replica trick

Kitaev Model: an Introduction

- Model Hamiltonian

$$H = - \sum_{\langle ij \rangle} J_{\alpha_{ij}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \quad ; \quad \begin{array}{c} \text{Diagram of three adjacent hexagons sharing a central vertex. The central vertex is labeled } z. \text{ The two vertices adjacent to } z \text{ are labeled } y \text{ and } x. \end{array}$$

- Majorana fermion (parton) representation: $b^x, b^y, b^z, c \Rightarrow \sigma^\mu = ib^\mu c$
- Parton Hamiltonian: define $ib_i^{\alpha_{ij}} b_j^{\alpha_{ij}} = \hat{u}_{ij}$

$$H \rightarrow \sum_{\langle ij \rangle} i J_{\alpha_{ij}} \hat{u}_{ij} c_i c_j$$

- Gutzwiller projection

$$\mathcal{P}_G = \prod_i \left(\frac{1 + D_i}{2} \right) \quad , \quad D_i = b_i^x b_i^y b_i^z c_i \leftarrow -i \sigma_i^x \sigma_i^y \sigma_i^z$$

- Separability of the entanglement entropy (Yao and Qi, PRL105 080501)

$$S_A = S_A^F + S_A^G$$

Kitaev Model: Gauge Theory

In the Majorana fermion representation, the model is equivalent to a system of free Majorana fermions (matter field) coupled to a static $\mathbb{Z}_2$ gauge field

- Define the gauge field $\hat{u}_{ij} = e^{iA_{ij}}$, then the elementary $\mathbb{Z}_2$ gauge flux reads

$$e^{i\Phi} = \exp\left(i \oint_{\partial p} A\right) = \prod_{\langle ij \rangle \in \partial p} \hat{u}_{ij} = \prod_{\langle ij \rangle \in \partial p} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} =: W_p$$

- The electric field in the gauge field that flips the gauge field is $b_i^{\alpha_{ij}} \rightarrow e^{i\pi E_{ij}}$
because $b_i^{\alpha_{ij}} \hat{u}_{ij} b_i^{\alpha_{ij}} = -\hat{u}_{ij}$
- Gutzwiller projection as a Gauss law constraint: $D_i = b_i^x b_i^y b_i^z c_i = 1$. The ground state is

$$\prod_i \left(\frac{1 + D_i}{2} \right) |u\rangle |M\rangle \quad ; \quad \text{Fix the gauge: } \langle u | \hat{u}_{ij} | u \rangle = +1 \forall \langle ij \rangle$$

- Both definitions $b_i^{\alpha_{ij}} = e^{i\pi E_{ij}}$ and $b_j^{\alpha_{ij}} = e^{i\pi E_{ij}}$ are valid.
- The Gauss law is a physical constraint and cannot be broken.

Kitaev Model: Properties of Spin Correlation Functions

- Two-spin correlation functions are short-ranged (Baskaran, Mandal, and Shankar, PRL**98**, 247201)

$$\langle \sigma_i^\mu \sigma_j^\nu \rangle = \langle M | i c_i c_j | M \rangle \underbrace{\langle u | i b_i^\mu b_j^\nu | u \rangle}_{\text{vanishes if } i b_i^\mu b_j^\nu \neq \hat{u}_{ij}}$$

- Multi-spin correlation functions can ONLY be string-like

$$\left\langle \prod_{\langle ij \rangle \in \mathcal{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right\rangle \equiv \langle \Sigma_{\mathcal{D}} \rangle$$

- $\mathbb{Z}_2$ algebra of strings

$$\mathcal{D} \ominus \mathcal{D}' := \mathcal{D} \cup \mathcal{D}' - \mathcal{D} \cap \mathcal{D}' \quad : \quad \left(\text{Diagram 1} \right) \ominus \left(\text{Diagram 2} \right) = \left(\text{Diagram 3} \right)$$

- String operators form a projective representation

$$\boxed{\Sigma_{\mathcal{D}} \Sigma_{\mathcal{D}'} = \varphi(\mathcal{D}, \mathcal{D}') \Sigma_{\mathcal{D} \ominus \mathcal{D}'}} \quad ; \quad \varphi(\mathcal{D}, \mathcal{D}') = \pm 1, \quad \varphi(\mathcal{D}, \mathcal{D}) = (-1)^{|\mathcal{D}|}$$

Organization: Equivalence Relation

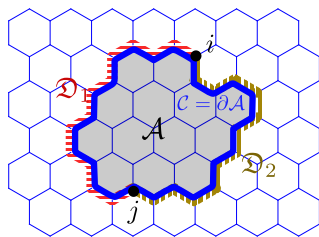
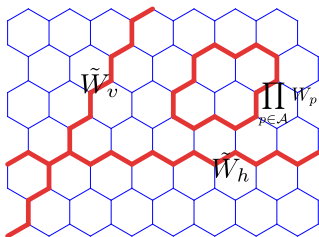
- Exact 1-form symmetry: Wilson loop $\Sigma_C |\psi_0\rangle = W_C |\psi_0\rangle = |\psi_0\rangle$
- The weights are identical up to a sign if the strings are identical up to a loop $\mathcal{C}$

$$\langle \Sigma_{\mathfrak{D} \ominus \mathcal{C}} \rangle = \varphi(\mathfrak{D}, \mathcal{C}) \langle \Sigma_{\mathfrak{D}} \Sigma_{\mathcal{C}} \rangle = \varphi(\mathfrak{D}, \mathcal{C}) \langle \Sigma_{\mathfrak{D}} W_{\mathcal{C}} \rangle = \varphi(\mathfrak{D}, \partial A) \langle \Sigma_{\mathfrak{D}} \rangle$$

- Equivalence relation:

$$[\mathfrak{D}] = \{ \mathfrak{D}' : \exists \mathcal{C} \text{ composed by closed loops such that } \mathfrak{D}' \ominus \mathcal{C} = \mathfrak{D} \}$$

- Endpoints perspective: $\text{EP}(\mathfrak{D}) = \text{EP}(\mathfrak{D}')$ iff $[\mathfrak{D}] = [\mathfrak{D}'] \Rightarrow$ Emerging fermions at endpoints of strings



Organization of Density Matrix

- Suppose the observables are a set of $\mathbb{Z}_2$ traceless Hermitian operators: $\{\mathcal{O}_n^2 = \mathbb{1}\}$, then the density matrix can be expanded in terms of these observables

$$\rho \propto \sum_n \text{Tr}(\mathcal{O}_n^\dagger \rho) \mathcal{O}_n = \sum_n \langle \mathcal{O}_n \rangle \mathcal{O}_n$$

- For the spin-1/2 Kitaev model, the observables are Pauli strings.

$$\rho \propto \sum_{\mathfrak{D}} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}}$$

- Organization due to the exact 1-form Wilson symmetry:

$$\rho \propto \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}} \quad ; \quad \mathcal{P}^{(p)} := \underbrace{\left(\frac{\mathbb{1} + \tilde{W}_h}{2} \right) \left(\frac{\mathbb{1} + \tilde{W}_v}{2} \right)}_{\text{topological projector (global Wilson)}} \underbrace{\prod_p \left(\frac{\mathbb{1} + W_p}{2} \right)}_{\text{flux-free projector}}$$

- Normalize $\text{Tr} \rho = 1$ to get the final result:

$$\rho = \frac{1}{2^{N-N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}}$$

Organization of Reduced Density Matrix

- Following the same spirit to construct the reduced density matrix:

- (i) Hermiticity
- (ii) Ability of reproducing expectation values of observables
- (iii) Normalization condition $\text{Tr} \rho = 1$

$$\rho_A \propto \sum_{\mathfrak{D} \subset A} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}} \propto \mathcal{P}_A^{(p)} \sum_{[\mathfrak{D}] \subset A} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}} \xrightarrow{\text{normalize}} \frac{1}{2^{N^A - N_p^A}} \mathcal{P}_A^{(p)} \sum_{[\mathfrak{D}] \subset A} \langle \Sigma_{\mathfrak{D}} \rangle \Sigma_{\mathfrak{D}}$$

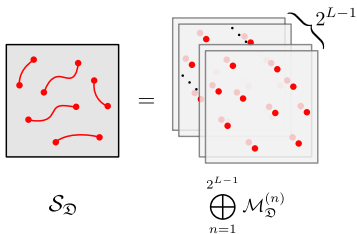
- Define *projected string operators*: $\mathcal{S}_{\mathfrak{D}}^{(A)} := \mathcal{P}_A^{(p)} \Sigma_{\mathfrak{D}}$. In this way, $\mathcal{S}_{\mathfrak{D}}^{(A)} = \mathcal{S}_{\mathfrak{D}'}^{(A)}$ if $\mathfrak{D} \sim \mathfrak{D}'$.
- Representing density matrices in terms of projected string operators:

$$\rho_A = \frac{1}{2^{N^A - N_p^A}} \sum_{[\mathfrak{D}] \subset A} \langle \Sigma_{\mathfrak{D}} \rangle \mathcal{S}_{\mathfrak{D}}^{(A)}$$

Remark: Since $\text{EP}(\mathfrak{D}) = \text{EP}(\mathfrak{D}')$ iff $[\mathfrak{D}] = [\mathfrak{D}']$, the equivalent fermion density matrix ρ_A^F looks the same as ρ_A .

$$\rho_A^F = \frac{1}{2^{N^A/2}} \sum_{\text{EP}(\mathfrak{D}) \subset A} (\text{fermion Op.}) = \frac{1}{2^{N^A/2}} \sum_{[\mathfrak{D}] \subset A} (\text{fermion Op.})$$

Organization: Block-Diagonal Structure



- Majorana representation of string operators: $\mathcal{M}_{\mathfrak{D}} := \prod_{\langle ij \rangle \in \mathfrak{D}} i c_j c_i$, which satisfies the same algebra (statistics)

$$\mathcal{M}_{\mathfrak{D}} \mathcal{M}_{\mathfrak{D}'} = \varphi(\mathfrak{D}, \mathfrak{D}') \mathcal{M}_{\mathfrak{D} \oplus \mathfrak{D}'}$$

- Difference in the Hilbert space dimension:

$$\mathrm{Tr}_{A,F} \mathcal{M}_{\emptyset}^{(A)} = 2^{N^A/2} \neq \mathrm{Tr}_A \mathcal{S}_{\emptyset}^{(A)} = 2^{N^A - N_p^A}$$

- Conjecture: projected string operators are *block diagonal*: $s_{\mathfrak{D}}^{(A)} = \frac{1}{2^{L-1}} \oplus_{n=1}^{2^{L-1}} \mathcal{M}_{\mathfrak{D}}^{(n)}$

Observation: A degenerate entanglement spectrum leads to the separability of the entanglement entropy

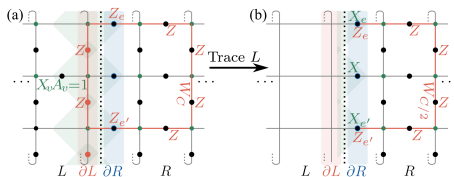
$$\rho = \frac{1}{d} \bigoplus_{n=1}^d \varrho \Rightarrow S^{(n)}(\rho) = \frac{1}{1-n} \ln \left[\frac{1}{d^{n-1}} \text{Tr}(\varrho^n) \right] = \underbrace{S^{(n)}(\varrho)}_{\text{matter}} + \underbrace{\ln d}_{\text{gauge}}$$

Questions:

- Can we directly prove that the entanglement spectrum is degenerate?
- How could they be related to *gauge fields*?

Exact Gauss Law $\times$ Exact 1-Form Wilson Symmetry: –Exact Block-Diagonal Structure

Review the method discovered by a recent work [W.-T. Xu *et al.* PRX **15**, 011001 (2025)]



- Exact Gauss law $[\rho, X_e] = 0 \forall e \in \partial R$: block-diagonal structure $\rho = \bigoplus_{\{x\}} \varrho_{\{x\}}$
- Exact 1-form Wilson loop symmetry $[W_{C/2}, \rho] = 0$ relates different blocks since $W_{C/2} X_e W_{C/2}^\dagger = -X_e \forall e \in \partial R$
- Yielding $2^{|\partial R|-1}$ -fold degeneracy $\rho = \frac{1}{2^{|\partial R|-1}} \varrho^{\oplus 2^{|\partial R|-1}}$

Block-Diagonal Density Matrix in the Kitaev Model

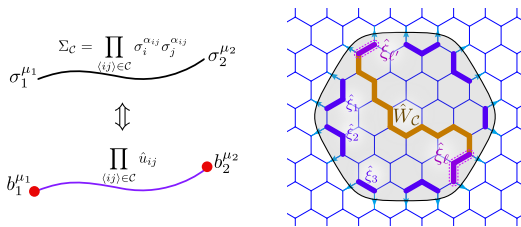
In the Kitaev model:

- Exact Gauss law: From the electric field $b_i^{\alpha ij} = e^{i\pi E_{ij}}$ at the boundary.
- Exact 1-form Wilson symmetry: $\langle W_C \rangle = \langle \Sigma_C \rangle = 1$ for any closed loop C .

But the gauge field here is *fictitious*. To use spin representation (physical):

- Connecting two *deconfined* degrees of freedom to form physical objects
- Use electric fields to define a set of operators $\hat{\xi}_n := \sigma_n^{\mu_n} \Sigma_{C_n} \sigma_{n+1}^{\mu_{n+1}}$ along the boundary of the entanglement cut.
- Eventually, we obtain $L = \frac{1}{2} |\partial A|$ independent $\hat{\xi}$ -operators, and they satisfy

$$[\mathcal{S}_{\mathfrak{D}}^{(A)}, \hat{\xi}_n] = 0 \quad ; \quad \mathfrak{D} \subset A$$



Block-Diagonal Density Matrix in the Kitaev Model

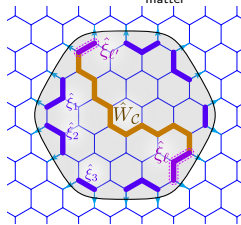
- String operators are block-diagonalized by $\hat{\xi}$ -operators
- Non-contractible Wilson operators flip $\hat{\xi}$ -operators, making different blocks identical
- Matching the Hilbert space dimension and considering the identical algebra between $\mathcal{S}$ and $\mathcal{M}$

$$\mathcal{S}^{(A)}_{\mathcal{D}} = \frac{1}{2^{L-1}} \bigoplus_{n=1}^{2^{L-1}} \mathcal{M}^{(n)}_{\mathcal{D}}$$

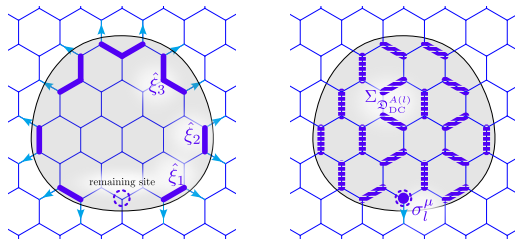
- Reduced density matrix is also block-diagonal: extensive degeneracy of the entanglement spectrum

$$\rho_A = \frac{1}{2^{L-1}} \bigoplus_{n=1}^{2^{L-1}} \rho_A^F \Rightarrow S_A^{(n)} = \frac{1}{1-n} \ln \text{Tr} \rho_A^n = \underbrace{S^{(n)}(\rho_A^F)}_{\text{matter}} + \underbrace{(|\partial A| - 1) \ln 2}_{\text{gauge}}$$

$$\begin{aligned} \Sigma_C &= \prod_{\langle ij \rangle \in C} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \\ \sigma_1^{\mu_1} &\text{---} \sigma_2^{\mu_2} \\ &\Updownarrow \\ b_1^{\mu_1} &\text{---} \prod_{\langle ij \rangle \in C} \hat{u}_{ij} \text{---} b_2^{\mu_2} \end{aligned}$$



Odd Number of Sites



- When N^A is odd, an equivalent fermion system is ill-defined: We need to borrow a fermion from the gauge sector.
- For odd N^A , there is always an unpaired site at the boundary.
- We can define an additional operator Ξ_A that also commutes with ρ_A , yielding an additional 2-fold degeneracy in the entanglement spectrum

$$\hat{\Xi}_A \equiv \sigma_l^\mu \Sigma_{D_{DC}}^{A(l)} \rightarrow \left(\prod_{\langle ij \rangle \in D_{DC}^{A(l)}} \hat{u}_{ij} \right) \underbrace{\left(b_l^\mu \prod_{i \in A} c_i \right)}_{\text{fermion parity}}$$

Summary

- The entanglement structure of the Kitaev ground state is studied completely in terms of **physical spin representation**.
- The exact **1-form Wilson symmetry** makes the density matrix **organized**. Such an organized form is equivalent to the fermion density matrix.
 - The **string operators** directly reflect the **fermionic statistics**.
 - The **gauge** DoF is encoded in the flux-free projector $\mathcal{P}^{(p)}$, and the **matter** DoF emerges at endpoints of projected string operators $\mathcal{S}_{\mathcal{D}}$.
- As proposed by [W.-T. Xu *et al.* PRX **15**, 011001 (2025)], the **exact 1-form Wilson symmetry** with the **exact Gauss law** gives an extensively degenerate entanglement spectrum.
 - *Glue* the fictitious electric fields back to form physical operators $\hat{\xi}$ in the spin representation.
 - The reduced density matrix is composed of identical copies of the fermionic density matrix.
 - Separability of the entanglement entropy
- **Odd** number of lattice sites: **Fermion parity** provides an additional degeneracy.

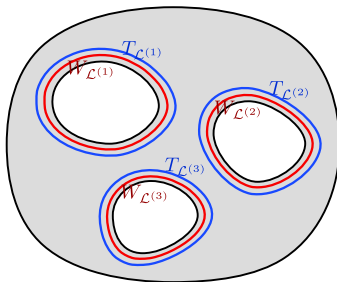
Unsolved Problems and Outlooks

- To get the **true topological entanglement entropy**, we need an exact 't Hooft symmetry and make it spontaneously broken.
- Higher spin Kitaev model: Expectation values don't provide sufficient conditions.
- Multi-connected subregion

Toric Code Model: True Topological Entanglement entropy

- Reduced density matrix of a Kitaev ground state

$$\rho_A = \frac{1}{2^{N^A - N_P^A - N_S^A - 2g}} \prod_{n=1}^g \left(\frac{\mathbb{1} + T_{\mathcal{L}(n)}}{2} \right) \prod_{n=1}^g \left(\frac{\mathbb{1} + W_{\mathcal{L}(n)}}{2} \right) \prod_{p \in A} \left(\frac{\mathbb{1} + P_p}{2} \right) \prod_{s \in A} \left(\frac{\mathbb{1} + S_s}{2} \right)$$

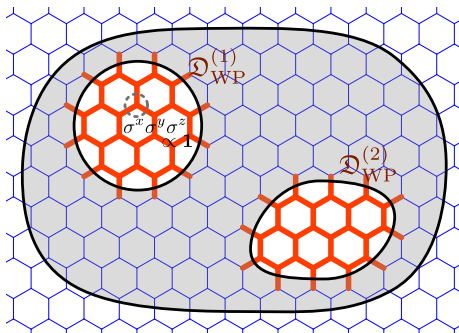


- Entanglement entropy: If the density matrix is proportional to a projector
 $\rho_A = \mathcal{CP}$

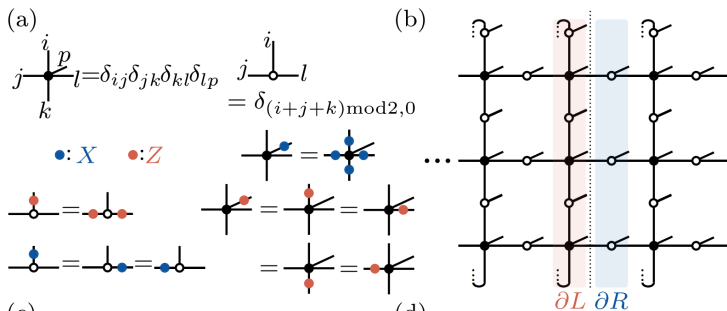
$$S_A^{(n)} = \frac{1}{1-n} \ln \text{Tr} \rho_A^n = \frac{1}{1-n} \ln \left(C^{n-1} \text{Tr} \rho_A \right) = -\ln C = (|\partial A| - (j+g)) \ln 2$$

Frame Title

- For a multi-connected region with genus number g , there is a set of *web patch* configurations $\{\mathfrak{D}_{\text{WP}}^{(1)}, \mathfrak{D}_{\text{WP}}^{(2)}, \dots, \mathfrak{D}_{\text{WP}}^{(g)}\}$ such that $\text{Tr}_{\bar{A}} \left(\Sigma_{\mathfrak{D} \ominus \mathfrak{D}_{\text{WP}}^{(n)}} \right) \neq 0$ if $\mathfrak{D} \subset A$
- In general, however, $\Sigma_{\mathfrak{D} \ominus \mathfrak{D}_{\text{WP}}^{(n)}} \neq \Sigma_{\mathfrak{D}} \Sigma_{\mathfrak{D}_{\text{WP}}^{(n)}}$, thus the reduced density matrix cannot be organized as before.
- The web patch is invalid in the fermion representation: $\text{Tr}_{\bar{A}, F} \left(\mathcal{M}_{\mathfrak{D} \ominus \mathfrak{D}_{\text{WP}}^{(n)}} \right) = 0$

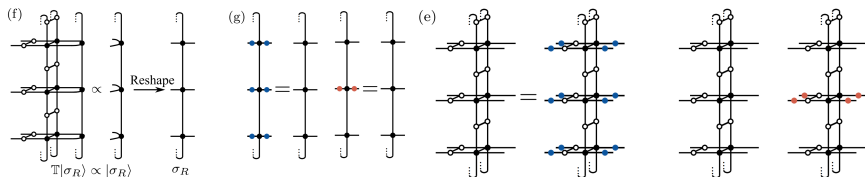


Contribution 2: Exact 1-Form Wilson



- **Symmetry of the tensor network** derives the **0-form Ising symmetry** and the **1-form Wilson loop symmetry**.

Contribution 2: Exact 1-Form Wilson



- Transfer matrix of a tensor network gives reduced density matrix (in the virtual space)

$$\mathbb{T}|\sigma_R\rangle \propto |\sigma_R\rangle \Rightarrow |\sigma_R\rangle \rightarrow \sigma_L \Rightarrow \rho \sim \sigma_L^T \sigma_R$$

- Symmetry of the tensor network becomes the symmetry of the reduced density matrix

$$\left(\prod_i X_i \otimes \prod_i X_i \right) \mathbb{T} \left(\prod_i X_i \otimes \prod_i X_i \right) = \mathbb{T} \Rightarrow \left(\prod_i X_i \right) \sigma_{L(R)} \left(\prod_i X_i \right) = \sigma_{L(R)}$$

$$(Z_j \otimes Z_j) \mathbb{T} (Z_j \otimes Z_j) = \mathbb{T} \forall j \quad Z_j \sigma_{L(R)} Z_j = \sigma_{L(R)} \forall j$$

- Therefore, two operators $\prod_i X_i$ and Z_j commute with the reduced density matrix, yielding 2-fold degeneracy in the entanglement spectrum.

Contribution 3: Spontaneous 1-form Symmetry Breaking

- Implementing the 1-form 't Hooft loop symmetry

$$\rho \sim \frac{1 + T_{C_y}}{2} \exp(-H_E)$$

- The 1-form 't Hooft loop symmetry is spontaneously broken: $T_{C_y} |\psi\rangle \neq |\psi\rangle$, so the expectation value satisfies the perimeter law: $\langle T_C \rangle \sim e^{-|C|}$.
- Entanglement entropy:

$$S_n = \frac{1}{1-n} \ln \text{Tr} \rho^n \stackrel{|\partial R| \rightarrow \infty}{\approx} - \frac{f(1/n) - f(1)}{1/n - 1} |\partial R| \underbrace{- \ln 2}_{\text{topo EE}}$$